

TSIO Innovation Hub MVP

Product Intent and Core Requirements

Audience	Pivota delivery team
Purpose	Product contract for discovery, product decomposition, implementation readiness, and MVP acceptance
Status	Working draft
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Requirements language. “Must” identifies an MVP requirement or release condition. “Should” identifies a preferred behavior that may be refined during discovery. “May” identifies an optional implementation choice. Product requirements define required outcomes; functional and technical specifications define the detailed implementation contract.

1. Product Intent

TSIO Innovation & Research produces proofs of concept, lessons learned, technical findings, demonstrations, reference architectures, code, and decision recommendations. Much of this work is valuable beyond the project that produced it, but it is currently difficult for other offices to discover and use.

The problem is not that the work has not been documented. The problem is that it is distributed across project folders, SharePoint locations, code repositories, videos, and individual team knowledge. A person must often already know the project name, the original team, or the document location before they can find anything useful.

The TSIO Innovation Hub should provide a single, governed entry point where a Judiciary stakeholder can:

- Discover innovation work relevant to a mission problem
- Understand what was tested and learned
- Distinguish an idea from a POC, pilot, or production pattern
- See the risks, constraints, and readiness of the work
- Locate authoritative supporting artifacts
- Understand what another office may reuse
- Identify the original contributors and current owners
- Contact I&R to discuss a use case, request a demonstration, or explore adoption

The Hub is not simply a document library. It is an engagement and transition mechanism.

Its purpose is to convert scattered innovation outputs into discoverable, understandable, and actionable institutional knowledge.

The Innovation Hub MVP must prove that existing I&R work can be curated into a useful product experience without replacing the repositories that currently hold the authoritative documents, code, and media.

2. Product Hypothesis

We believe that stakeholders are more likely to engage with, reuse, and adopt innovation work when it is presented through a consistent record that explains:

- The mission problem
- The users affected
- What was explored
- What was demonstrated
- What was learned
- How mature the work is
- What review has occurred
- What can be reused
- What remains unresolved
- Who to contact
- What action can be taken next

The Innovation Hub MVP should test whether this presentation materially improves:

- Awareness of I&R capabilities
- Discovery of prior work
- Understanding of POC outcomes
- Reuse of lessons and technical patterns
- Engagement between I&R and operational offices
- Identification of potential adoption partners
- Visibility into I&R's contribution to downstream outcomes

MVP success will be evaluated by whether intended users can discover, understand, assess, reuse, and act on innovation work with materially less reliance on prior project knowledge or direct I&R assistance.

3. Primary Users and Jobs to Be Done

3.1 Decision-Maker

A TSIO or AO leader wants to understand whether an innovation effort is relevant, credible, and worth further investment.

- What problem does this address?
- Why does it matter?
- What evidence was produced?
- What are the main benefits, risks, and constraints?
- How mature is the work?
- What decision or sponsorship is needed?
- Who would own the next step?

Desired outcome: Determine whether the effort is relevant and credible enough to justify sponsorship, additional evaluation, adoption exploration, or no further action.

Authorized product actions: Search or browse; view executive perspective; review evidence, maturity, risk, ownership, and next step; open permitted artifacts; initiate an engagement request.

3.2 Operational Leader

An application owner, service owner, program manager, or branch chief wants to know whether the work could address a real operational need.

- Does this apply to my office or workflow?
- What would adoption require?
- What responsibilities would my team assume?
- What dependencies, skills, or infrastructure are needed?
- Is I&R looking for an adopter or collaborator?
- How do I begin a discussion?

Desired outcome: Determine whether the work applies to an operational need and what organizational, technical, or ownership commitments would be required to advance it.

Authorized product actions: Search or browse; view applicability and reuse guidance; review ownership and dependencies; request a demonstration, adoption discussion, or related-problem discussion.

3.3 Technical Adopter

An architect, developer, cloud engineer, security engineer, or DevSecOps practitioner wants to evaluate the work for technical reuse.

- How was the POC designed?
- What tools and services were used?
- What code or infrastructure assets are available?
- What technical decisions were made?
- What limitations were discovered?
- What is suitable for reuse?
- What must change before production?
- What security and data constraints apply?
- Where are the authoritative technical artifacts?

Desired outcome: Determine what is technically reusable, what remains uncertain, and what must change before use in a production or office-specific environment.

Authorized product actions: View technical perspective; inspect architecture, findings, limitations, dependencies, and permitted source artifacts; request technical guidance or an adoption discussion.

3.4 Innovation Contributor

A court, AO office, or technical team wants to share an idea, experiment, implementation, or lesson learned.

- Describe the problem and work completed
- Identify the contributing office and people
- Retain attribution
- Indicate maturity and ownership
- Link existing artifacts
- Submit the work for curation without implying formal approval

Desired outcome: Submit useful innovation work or a lesson learned while preserving attribution, ownership, maturity context, and the distinction between contribution and endorsement.

Authorized product actions: Submit existing innovation work; provide source links, owners, contributors, limitations, and collaboration preferences; receive a clear confirmation of the curation process.

3.5 I&R Curator

An authorized I&R team member manages the quality and lifecycle of Hub content.

- Create and edit innovation records
- Summarize source material
- Apply consistent metadata
- Assign maturity and review labels
- Link authoritative artifacts
- Preserve contribution history
- Publish and unpublish records
- Track review dates
- Archive, supersede, or retire stale content
- Review engagement associated with each record

Desired outcome: Maintain a trustworthy, current, consistently structured body of innovation knowledge and route meaningful stakeholder engagement.

Authorized product actions: Create, edit, review, publish, unpublish, supersede, archive, or retire records; manage metadata and artifacts; review submissions and engagement; manage approved configuration.

4. Core MVP Experience and User Journeys

The MVP must center on one primary experience: a user arrives with a mission problem or area of interest, discovers relevant innovation work, understands its status and applicability, and takes an informed next step.

4.1 Primary Journey — Discover and Act on Existing Innovation

1. The user arrives with a mission problem, operational need, technology question, or area of interest.
2. The user searches or browses by problem, mission area, technology, contributing office, maturity, review status, or keyword.
3. The user sees clearly labeled innovation records and identifies one or more relevant records.
4. The user opens a record and immediately understands the problem, outcome, maturity, review status, primary findings, ownership, and recommended next step.
5. The user uses the executive or technical perspective as appropriate without creating a duplicate record or conflicting version of the evidence.
6. The user opens authoritative supporting artifacts where additional evidence is needed and access is permitted.
7. The user determines whether the work is relevant for reuse, further discussion, demonstration, adoption exploration, or no action.
8. The user takes an action such as requesting a demo, discussing a related problem, requesting technical guidance, exploring adoption, sharing related work, or contacting I&R.
9. The request is associated with the originating context and routed to I&R with enough information for follow-up.

The MVP must optimize for this journey before adding portfolio management, collaboration spaces, social features, or advanced automation.

4.2 Secondary Journey — Submit an Opportunity

1. A user describes a mission problem or workflow friction rather than beginning with a requested application.
2. The user identifies who is affected, the current workflow, impact, desired outcome, constraints, related attempts, office, and discovery participants.
3. The user submits the opportunity and receives confirmation that submission does not imply I&R acceptance.
4. The submission enters an I&R review process and remains traceable to its originating office and contact.

4.3 Secondary Journey — Share Existing Innovation

1. A contributor describes the problem addressed, work completed, contributing office, maturity, current owner, artifacts, limitations, and collaboration preference.
2. The contribution is submitted with attribution intact.
3. The contribution enters a curation workflow before any publication.
4. If published, the resulting innovation record remains linked to its source contribution and authoritative artifacts.

4.4 Secondary Journey — Curate and Govern Content

1. An authorized curator creates a record from source material or an accepted contribution.
2. The curator applies the content model, metadata, maturity, review information, attribution, source links, and next action.
3. The record is reviewed against publication requirements.
4. The record is published, updated, superseded, archived, retired, or unpublished through a governed lifecycle.
5. Material changes and review dates remain auditable.

5. MVP Scope and Feature Decomposition

Each feature group below is an independently traceable part of the MVP. A partial implementation of one capability must not be treated as completion of another. Detailed fields, validation rules, error behavior, API design, and persistence design belong in the downstream functional and technical specifications.

5.1 F1 — Innovation Catalog

- **F1.1** — Provide a browsable catalog of curated innovation records.
- **F1.2** — Each catalog card must show the title and a one-sentence problem or outcome.
- **F1.3** — Each card must visibly communicate technology or capability area, maturity, review status, and contributing office.
- **F1.4** — Each card must show an appropriate reuse or engagement indicator when one is configured, such as available for demonstration, seeking adoption partner, technical playbook available, reference pattern available, monitoring only, or archived.
- **F1.5** — Each card must show last-reviewed date and current lifecycle state where that state affects interpretation.
- **F1.6** — The catalog must not visually imply that all records are equally mature, approved, current, or reusable.

Completion criterion: A user can browse multiple records and distinguish relevance, maturity, review status, contributor context, and actionability without opening every record.

5.2 F2 — Search and Discovery

- **F2.1** — Support problem-oriented search so users do not need to know internal project names or repository locations.
- **F2.2** — Search must cover record titles, problem statements, summaries, findings, tags, mission areas, technology areas, and artifact names where permitted.
- **F2.3** — Support filtering by mission or business area, problem type, technology, maturity, review status, contributing office, reuse potential, artifact availability, and active/superseded/retired state.
- **F2.4** — Search and filter behavior must preserve visible trust information in results.
- **F2.5** — A query expressed in user problem language, such as “protect court audio,” must be able to surface relevant work without requiring the formal project title when the content and metadata support that relationship.

Completion criterion: A user can locate relevant innovation work from problem-oriented language and refine results using product metadata.

5.3 F3 — Innovation Record

- **F3.1 Problem and Context** — Explain the mission or operational problem, affected users, current workflow or constraint, and why experimentation was appropriate.
- **F3.2 What Was Explored** — Explain the hypothesis, capability, or approach tested; scope boundaries; and technologies or methods used.
- **F3.3 Outcome and Evidence** — Explain what was demonstrated, what evidence was produced, what worked, what did not, what uncertainty was reduced, and what decision the work enabled.
- **F3.4 Key Findings** — Surface reusable architectural, security, cloud/platform, performance, user-experience, data, testing, operational, cost, or scalability findings where applicable.
- **F3.5 Maturity and Readiness** — Show current maturity, review status, what the work is ready for, what it is not ready for, and what is required before the next stage.
- **F3.6 Reuse Guidance** — State what another office can reuse, what should be adapted, what should not be copied directly, what assumptions are environment-specific, and what skills/services/dependencies are required.
- **F3.7 Ownership and Attribution** — Identify opportunity source, contributing office, I&R contribution, technical contributors, current owner, and operational or production owner where applicable.
- **F3.8 Authoritative Artifacts** — Link to authoritative lessons learned, POC reports, decision briefs, diagrams, demo videos, repositories, infrastructure definitions, test results, security findings, and technical playbooks as available and permitted.
- **F3.9 Next Action** — Provide only appropriate next actions for the record, including request a demonstration, discuss a related use case, explore adoption, request technical guidance, share related work, or contact I&R.

Completion criterion: From one coherent record, the user can understand the problem, evidence, maturity, review context, reuse potential, ownership, source artifacts, and appropriate next action.

5.4 F4 — Executive and Technical Perspectives

- **F4.1** — One innovation record must support both executive and technical perspectives without creating duplicate source records.
- **F4.2** — The executive perspective should prioritize mission problem, strategic relevance, outcome, evidence, benefits, risks, maturity, decision recommendation, ownership, and next step.
- **F4.3** — The technical perspective should prioritize architecture, tools/services, data flow, security considerations, testing, known limitations, source artifacts, production-readiness gaps, reuse guidance, and dependencies.
- **F4.4** — Both perspectives must remain grounded in the same underlying evidence, maturity, review status, ownership, and authoritative artifacts.

Completion criterion: An executive and a technical adopter can obtain audience-appropriate information without encountering conflicting versions of the underlying record.

5.5 F5 — Existing Lessons-Learned Content

The MVP must explicitly support lessons-learned documents already produced through the I&R POC process. The goal is to curate and link, not to rewrite or relocate every authoritative document.

- **F5.1** — Treat the existing lessons-learned document or other authoritative source as the source of record.
- **F5.2** — Create a structured innovation record around the source and extract the most important reusable findings.
- **F5.3** — Apply metadata, maturity, review, ownership, attribution, and review-date information.

- **F5.4** — Link back to the authoritative source and make the record discoverable through problem-oriented search.
- **F5.5** — Use the Audio Security POC as a strong initial candidate because its architectural, security, performance, cloud-environment, testing, and production-readiness findings can exercise the full content model.

Completion criterion: A user can discover and understand the reusable value of an existing lessons-learned artifact without first knowing where the original document is stored or reading it in full.

5.6 F6 — Opportunity Submission

- **F6.1** — Provide a structured submission flow that begins with the problem or workflow friction, not a requested application.
- **F6.2** — Capture who is affected, how the work is performed today, impact, desired outcome, known constraints, related work attempted, submitting office, and participants available for discovery.
- **F6.3** — Allow the submitter to characterize the request as a current mission problem, emerging-technology question, request for research, potential POC, request for demonstration, collaboration opportunity, or existing innovation work to share.
- **F6.4** — Clearly state that submission does not imply acceptance into the I&R portfolio.
- **F6.5** — Record the submission so authorized I&R users can review and disposition it through the defined workflow.

Completion criterion: A user can submit a problem with sufficient context for I&R to determine fit, relevance, research value, feasibility, and capacity without the submission implying approval.

5.7 F7 — Share Existing Innovation Work

- **F7.1** — Provide a separate contribution flow for teams that already have innovation work to share.
- **F7.2** — Capture the problem addressed, description of the work, contributing office, current maturity, current owner, available artifacts, known limitations, contact person, and whether collaboration or reuse is encouraged.
- **F7.3** — Preserve contributor attribution and current ownership through the curation process.
- **F7.4** — Require curation before publication and avoid language that implies central endorsement merely because a contribution was submitted.

Completion criterion: Existing innovation work can enter the Hub through a governed contribution path without losing attribution or being presented as approved before curation.

5.8 F8 — Engagement Routing

- **F8.1** — Provide record-level and general calls to action for demonstration, related-use-case discussion, adoption exploration, technical guidance, sharing related work, or contact with I&R.
- **F8.2** — Capture request type, originating innovation record where applicable, user name, office, contact information, description of the need, and desired next step.
- **F8.3** — Allow the first MVP implementation to use email for the conversation while separately recording that the engagement action occurred.
- **F8.4** — The routing destination must be configurable by authorized users or approved configuration without requiring a code change or application redeployment.
- **F8.5** — The configured display language should direct users to the TSIO Innovation & Research team. The initial shared address is AOml_TSO_IRB_Team@ao.uscourts.gov unless changed through approved configuration.

- **F8.6** — Suggested subject patterns may include Innovation Opportunity – [Office] – [Topic], Demo Request – [Innovation Record], Adoption Discussion – [Innovation Record], and Technical Guidance – [Innovation Record].

Completion criterion: User interest becomes a traceable engagement associated with the correct context and routed to the designated I&R channel.

5.9 F9 — Curation and Administration

- **F9.1 Curator Summary** — Provide an authorized summary view of records, submissions, engagements, review needs, and other attention items using live product data.
- **F9.2 Record Management List** — Provide an authorized, filterable view of innovation records across lifecycle states.
- **F9.3 Record Creation** — Allow an authorized curator to create a complete innovation record from source material or an accepted contribution.
- **F9.4 Record Editing** — Allow authorized editing of all product-required record content and metadata.
- **F9.5 Artifact Management** — Allow authorized users to add, update, and remove authoritative artifact links without creating uncontrolled copies.
- **F9.6 Maturity Management** — Allow authorized assignment and update of maturity while preserving the history of material changes.
- **F9.7 Review Status Management** — Allow authorized management of applicable review statuses independently from maturity.
- **F9.8 Attribution and Ownership** — Allow maintenance of contributing offices, contributors, I&R contribution, current steward, operational owner, and production owner where applicable.
- **F9.9 Publication Lifecycle** — Support draft, review, publish/unpublish, supersede, archive, and retire behaviors consistent with the lifecycle rules in this PRD.
- **F9.10 Publication Gate** — Prevent publication when required product information is absent, including problem statement, owner/steward, maturity, review status, attribution, source basis, last-reviewed date, and applicable disclaimer.
- **F9.11 Audit History** — Provide a chronological history of material content, governance, and lifecycle changes.
- **F9.12 Opportunity Submission Queue** — Allow authorized I&R users to review and disposition opportunity submissions.
- **F9.13 Contribution Submission Queue** — Allow authorized I&R users to review, disposition, and where appropriate initiate record creation from innovation contributions.
- **F9.14 Engagement Activity** — Allow authorized I&R users to review engagement activity and record an appropriate follow-up status.
- **F9.15 Settings Management** — Allow authorized management of approved configurable settings, including engagement-routing destination.
- **F9.16 Content Model Reference** — Provide authorized users with accessible definitions of maturity, review status, lifecycle, and publication requirements so governance is applied consistently.

Completion criterion: Authorized curators can govern the complete content and engagement lifecycle using product data, with publication gates and auditability that prevent incomplete or misleading records from being treated as complete.

6. Content, Maturity, Trust, and Lifecycle Model

6.1 Maturity

Idea

A problem, opportunity, or concept that has not yet been validated.

Evaluated Idea

An idea that has been reviewed for relevance, feasibility, and potential value.

Experiment / POC

A controlled effort that produced evidence and findings, but is not production-ready.

Prototype / Pilot

A capability tested with representative users, workflows, integrations, or environments.

Production / Validated Pattern

A deployed capability or reviewed approach with sufficient evidence to serve as a reference for reuse.

Archived / Retired

Work retained for institutional learning but no longer active or recommended.

6.2 Review Status

Review status must remain separate from maturity. A sophisticated POC may still lack security or policy review. A production example from another office may not be validated as a Judiciary-wide pattern.

- Submitted
- Curated for completeness
- Technically reviewed
- Security reviewed
- Policy reviewed
- Validated for reuse
- Superseded
- Retired

The product must make clear that:

- POC does not mean production-ready.
- Published does not mean approved for adoption.
- Community-submitted does not mean centrally endorsed.
- Validated for reuse does not eliminate local review requirements.

6.3 Lifecycle and Governance Rules

- Maturity and review status are independent and must not be automatically inferred from one another.
- Publication state is separate from maturity and review status.
- A published record must meet the publication gate defined in F9.10.
- A record may be unpublished when its content is no longer appropriate for stakeholder access or requires material correction.
- A superseded record should remain discoverable when it has institutional learning value, but it must clearly identify that it has been superseded and, where known, the successor record or artifact.
- A retired or archived record must not be presented as a recommended current pattern.
- Material lifecycle or governance changes must be captured in audit history.

- The product may support a workflow such as Draft → Submitted for Review → Published → Superseded / Archived / Retired, but the exact state persistence and implementation belong in downstream specifications.

7. Conceptual Information and Data Model

This section defines the product concepts and business relationships that implementation must preserve. It does not prescribe database table names, column names, DDL, API payloads, or persistence technology.

Product Concept	Business Meaning
Innovation Record	The structured representation of one idea, experiment, POC, pilot, validated pattern, production example, or archived learning item.
Artifact	An authoritative source associated with an innovation record, such as a lessons-learned document, report, architecture diagram, demo video, repository, test result, or security finding.
Contributor	A person, court, AO office, or technical team credited for the work.
Owner / Steward	The person or organizational entity responsible for the current accuracy, maintenance, or advancement of the record.
Opportunity Submission	A mission problem, research question, collaboration need, or other opportunity submitted to I&R for consideration.
Innovation Contribution	Existing innovation work submitted for curation and possible publication.
Engagement Request	A request for demonstration, guidance, discussion, adoption exploration, or other follow-up.
Maturity	The current developmental stage of the innovation work.
Review Status	A representation of governance or review that has occurred, maintained independently from maturity.
Mission / Technology Taxonomy	Controlled metadata used to support problem-oriented discovery and filtering.
Audit Event	A material change to product content, governance status, ownership, configuration, or lifecycle that requires traceability.

7.1 Required Business Relationships

- One innovation record may reference multiple authoritative artifacts.
- One innovation record may have multiple contributors and contributing offices.
- A published innovation record must have a current owner or steward.
- Maturity, review status, and publication state must remain distinct concepts.
- An engagement request may originate from an innovation record or from a general I&R contact action.
- An innovation contribution may result in an innovation record after curation but submission alone does not create an approved or published record.
- An opportunity submission must preserve its originating office/contact and curator disposition history where applicable.
- Audit history must remain associated with the governed object or setting that changed.

7.2 Core Innovation Record Information

- Title and unique identifier
- Short summary
- Problem statement
- Target users
- Mission area
- Technology areas
- Scope
- Hypothesis or objective
- Outcome
- Evidence
- Key findings
- Benefits
- Risks
- Limitations
- Maturity
- Review statuses
- Reuse potential
- Production-readiness gaps
- Recommended next step
- Contributing offices
- Contributors
- I&R contribution
- Current owner/steward
- Operational owner where applicable
- Production owner where applicable
- Artifact links
- Engagement actions
- Created date
- Last-reviewed date
- Next-review date
- Publication state
- Retirement or supersession reason where applicable

Detailed field definitions, required/optional rules, validation, exact state values, and logical/physical schema belong in the FRD and data specification.

8. Experience and Design Principles

Problem-Oriented Discovery

Users should not need to understand I&R's internal project naming or folder structures.

Evidence Before Promotion

The Hub should communicate findings and maturity honestly. It should not make every POC look like a success story.

One Record, Multiple Audiences

Executive and technical views should remain grounded in the same evidence.

Curate and Link

The Hub should summarize and organize authoritative content rather than creating uncontrolled copies.

Visible Trust

Maturity, review status, owner, and review date should be visible and understandable.

Actionable Pages

Every relevant record should help the user take a next step.

Attribution Without Centralization

The Hub should amplify work from across the Judiciary and preserve contributor credit without suggesting that I&R owns all innovation.

Maintainability Over Novelty

The MVP should favor a clear, supportable internal product over a visually elaborate but difficult-to-maintain portal.

Accessible by Default

Core product journeys must be usable by users with disabilities and conform to the approved accessibility baseline.

Governance in the Experience

Trust, ownership, review, lifecycle, and readiness information should be visible where it affects user interpretation or action rather than hidden only in administrative metadata.

9. Product-Level Non-Functional Requirements

The MVP must treat non-functional requirements as verifiable product requirements. Specific test tools may be selected by the delivery team, but the verification evidence must demonstrate the required outcome.

Category	Product Requirement	MVP Target	Verification Expectation
Accessibility	Core stakeholder and curator journeys conform to WCAG 2.1 AA unless a different Judiciary-approved baseline is established during discovery.	All MVP journeys	Automated and manual accessibility verification; unresolved critical issues block release.
Performance	Catalog, search, record viewing, and form interactions meet response-time targets agreed during discovery before implementation baseline.	Targets baselined before build	Repeatable performance verification against the agreed target.
Browser Compatibility	Core MVP journeys function on the Judiciary-approved browser set.	Browser list confirmed during discovery	Cross-browser validation on the approved set.
Reliability	User submissions and curator changes must not be silently lost and must produce clear success/failure feedback.	All write workflows	Functional and failure-path testing.
Auditability	Material governance, lifecycle, ownership, and configuration changes are traceable.	All governed actions	Audit-history verification.
Maintainability	The delivered solution can be understood, operated, configured, and extended by a receiving technical team.	MVP handoff	Independent handoff review using source, deployment, configuration, and operational documentation.
Security	Protected functions and restricted content are inaccessible without appropriate authorization and operational security controls are present.	All protected capabilities	Security verification and negative-path testing.
Traceability	Product requirements are traceable to functional design, implementation, and verification evidence.	All MVP requirements	RTM review before release.

10. Security, Privacy, and Access Requirements

Security requirements in this PRD define required product outcomes. The exact control implementation, framework, header configuration, environment-variable names, authentication protocol, and infrastructure mechanism belong in the technical and security specifications.

- **SEC-01** — Administrative and curation capabilities must require authorized access.
- **SEC-02** — Unauthorized or unauthenticated users must not gain access to protected functions or restricted content.
- **SEC-03** — Authentication and authorization decisions must be auditable where required for governance and investigation.
- **SEC-04** — Publication of an innovation record must not implicitly broaden access to a restricted artifact. Artifact access remains governed by the authoritative source system and applicable permissions.
- **SEC-05** — User-submitted contact and opportunity information must be handled according to applicable Judiciary privacy and data-protection requirements.
- **SEC-06** — Public or broadly accessible submission capabilities must include appropriate protection against automated abuse and excessive requests.
- **SEC-07** — Security-sensitive failure conditions must default to a protected state rather than allowing access or submission when a required security control is unavailable.
- **SEC-08** — Credentials, secrets, and sensitive operational configuration must not be embedded in committed application source code.
- **SEC-09** — Development or test-only access mechanisms must not be active in operational environments.
- **SEC-10** — Operational deployments must implement appropriate application and HTTP security protections consistent with the approved technical/security baseline.
- **SEC-11** — Security review status must remain distinguishable from general technical review and from maturity.
- **SEC-12** — Security and access decisions required for a capability must be resolved or explicitly classified before that capability is ready for implementation or deployment.

11. External Integrations and Dependency Decisions

Every unresolved integration or external dependency must be explicitly classified as an implementation blocker, an accepted assumption, or an approved temporary development stub. Temporary mechanisms must have a documented condition that prevents them from being treated as operationally complete.

Dependency	MVP Need	Current State	Temporary Approach	Operational Blocker?
Identity and access	Required for protected curator/admin workflows	TBD during discovery	Development-only access mechanism may be permitted if explicitly approved	Yes, before non-development deployment
Hosting environment	Required for deployment and operational controls	TBD during discovery	Local/development deployment permitted	Yes, before operational deployment
Engagement routing	Required for MVP actions	Shared I&R address identified; confirm	Email routing is acceptable for MVP if action is separately recorded	No, if approved address and routing are confirmed
Authoritative artifact repositories	Required to link evidence	Multiple existing repositories expected	Link to authoritative source; do not migrate by default	No, but initial source locations must be confirmed

Dependency	MVP Need	Current State	Temporary Approach	Operational Blocker?
Automated-submission protection	Required where exposure warrants it	Mechanism TBD	Controlled development bypass may be allowed	Yes where the approved operational security baseline requires the control
Usage and engagement analytics	Required to test approved success measures	Metrics and collection method TBD	Basic approved event tracking may be sufficient for MVP	Decision required before analytics implementation; not necessarily launch-blocking

11.1 Stub and Blocker Governance

- A temporary development stub must have explicit acceptance criteria and an explicit condition describing where it may not be used.
- A production blocker must remain visible in the decision register and release review until resolved.
- A stub must not silently become the operational implementation merely because the feature appears functional in development.
- The technical specification must document the approved mechanism, configuration, security behavior, and verification evidence for each resolved integration.

12. Initial Content and Launch Acceptance

The MVP should begin with a deliberately small content set that proves the content, trust, lifecycle, and engagement model. Launch content is a product acceptance condition, not an optional post-build activity.

Launch Condition	Minimum Requirement	Why It Matters
Published content threshold	At least 3 published innovation records	Demonstrates a usable catalog rather than an empty shell.
Technical reuse example	At least 1 published record with significant reusable technical findings and source artifacts	Exercises technical perspective, reuse guidance, and artifact linkage.
Executive decision example	At least 1 published record that supports an executive decision or sponsorship discussion	Exercises executive perspective and decision-oriented content.
Adoption/collaboration example	At least 1 published record seeking an adopter, collaborator, demonstration, or other concrete engagement	Exercises actionable next steps and routing.
Lifecycle transparency example	At least 1 archived, retired, stopped, or superseded experiment retained for institutional learning	Demonstrates that the Hub does not present every experiment as a success or current pattern.
Complete governance metadata	Every published record includes maturity, applicable review status, attribution, owner/steward, last-reviewed date, source basis/artifact, and appropriate next action	Ensures visible trust and publication integrity.

The Audio Security POC is a strong candidate for the initial set because it contains defense-in-depth findings, GPU/CPU service separation, Azure Government Cloud constraints, infrastructure state-management guidance, performance limitations, testing gaps, security recommendations, production-readiness requirements, and future development paths.

13. MVP Success Criteria and Acceptance Measures

MVP acceptance should be based on whether intended users can complete the product jobs defined in this PRD. Usage and engagement metrics that measure the broader product hypothesis must be selected during discovery and supported by an approved collection method.

User / Area	Acceptance Measure	Evidence
Decision-Maker	Can locate a relevant innovation record using mission/problem language without knowing the formal project name; can identify problem, outcome, evidence, maturity, review context, risks/constraints, owner, and recommended next step from the record.	Task-based usability review and requirement verification.
Operational Leader	Can determine whether a record may apply to an office/workflow, what adoption would require, who owns the next step, and how to initiate a relevant discussion.	Task-based usability review and call-to-action verification.
Technical Adopter	Can find technical perspective, authoritative artifacts, reusable elements, limitations, production-readiness gaps, security/data constraints, dependencies, and technical contact path.	Task-based technical-adopter review.
Contributor	Can submit an opportunity or existing innovation example, understand what happens next, and see that attribution/ownership information is captured without implying endorsement.	Submission workflow verification.
Curator	Can create, edit, review, publish, unpublish, supersede/retire, and update a complete record; manage submissions, engagement, required metadata, source links, and approved configuration; and review material audit history.	Curator acceptance testing across all F9 sub-features.
Engagement	A demo, discussion, adoption, guidance, or related request reaches the designated I&R channel with user/context information and is recorded with originating record and request type where applicable.	End-to-end engagement test.
Pivota Delivery	Requirements are traceable; user journeys and design decisions are reviewable; source code is standard and maintainable; primary workflows have automated tests; deployment/configuration instructions exist; major requirements/approvals are auditable; handoff can be understood by a separate technical team.	RTM, test, documentation, and handoff review.

14. Explicit MVP Boundaries

The MVP will not initially:

- Replace SharePoint or Git repositories
- Migrate every historical document
- Manage POC execution
- Provide enterprise portfolio management
- Automatically determine maturity or approval
- Replace architecture, security, legal, or policy review
- Deploy POCs into production
- Provide broad social networking or discussion forums
- Make autonomous investment decisions
- Publish sensitive material outside approved access boundaries
- Attempt to operationalize the entire Innovation & Research Pipeline

The MVP must prove discovery, understanding, trust, attribution, curation, lifecycle transparency, and engagement. Capabilities outside these boundaries require explicit product-owner approval and a traceable PRD scope change before inclusion in the MVP.

15. Specification Governance and Required Delivery Artifacts

This PRD is the product contract. It defines the product problem, scope, users, required behavior, governance, boundaries, product-level quality expectations, and acceptance conditions. It intentionally does not duplicate detailed functional, UX, API, data, or technical implementation specifications.

Artifact	Required Scope	Authority
PRD — Product Contract	Product intent, users, scope, feature outcomes, governance, boundaries, acceptance, release conditions	Governs what the product must accomplish and why.
FRD — Functional Contract	Detailed functional behavior, fields, validation, permissions, business rules, exceptions, and detailed acceptance criteria	Governs how product behavior works from the user/business perspective.
UX Specification — Experience Contract	Information architecture, journeys, screens, interaction patterns, responsive behavior, content hierarchy, and accessibility treatment	Governs the intended experience and interaction behavior.
Technical Architecture — Implementation Contract	Application architecture, integrations, authentication implementation, infrastructure, deployment, technical controls, and major technology choices	Governs how the solution is implemented and operated.
API and Data Specifications	Exact interface paths/methods/payloads, logical/physical schemas, canonical enum/state values, and persistence details where needed	Governs precise machine-level contracts.
Integration Specification	External system contracts, configuration, operational dependencies, stub criteria, and production blockers	Governs external dependency behavior.
RTM / Test Plan — Verification Contract	Requirement-to-design-to-test traceability, verification methods, test cases, results, and release evidence	Governs proof that requirements were implemented and verified.
Deployment / Configuration / Handoff Package	Deployment instructions, environment configuration, operational guidance, known limitations, support assumptions, and receiving-team information	Governs operational transfer and maintainability.

15.1 Required Implementation References

Before a feature is baselined for implementation, the delivery team must identify and consult the current approved specifications applicable to that feature. At minimum, implementation must be traceable to the PRD, applicable FRD requirements, applicable UX specification, applicable architecture/integration/data contracts, and the RTM/test plan.

15.2 Conflict and Change Rule

- Downstream specifications may add detail but must not silently change the product intent, scope, trust model, or acceptance conditions in this PRD.
- When governing documents conflict, implementers must not silently choose one interpretation.
- The conflict must be escalated to the appropriate product/technical decision owner, resolved, and reflected in the authoritative specifications.
- Material product changes require PRD update or documented product-owner disposition and must remain traceable through the RTM and change history.
- API paths, database column names, framework-specific security mechanisms, environment-variable names, and test-tool selections should not be duplicated into the PRD unless they are themselves customer-mandated product constraints.

16. Decisions Required During Pivota Discovery

Discovery decisions refine implementation without changing the core product intent. Each decision must have an owner, required-by point, and blocker classification before the affected implementation begins.

Decision	Needed By	Blocking?	Decision Owner
Initial hosting environment	Before deployment architecture is baselined	Yes	Product owner + technical authority
Identity and access approach	Before protected curator/admin implementation	Yes	Security/technical authority + product owner
Shared contact address	Before engagement routing is baselined	Yes for engagement release	I&R product owner
Initial POC content set	Before launch-content preparation	Yes for launch	I&R product owner / curator
Authoritative artifact locations	Before initial records are published	Yes for affected records	Content curator / source owner
Product owner	Before requirements baseline	Yes	TSIO/I&R leadership
Content curator	Before curation workflow acceptance	Yes	TSIO/I&R leadership
Publishing authority	Before publication workflow baseline	Yes	TSIO/I&R leadership
Required review roles	Before trust/governance workflow baseline	Yes	Product owner + governance stakeholders
Initial taxonomy	Before catalog/search baseline	Yes	Product owner + curator
Email routing sufficient for MVP engagement tracking?	Before engagement implementation	Yes for engagement design	Product owner
Usage and engagement metrics	Before analytics instrumentation	No for all development; required before metric collection	Product owner
Receiving technical team	Before handoff planning is finalized	Yes for handoff acceptance	TSIO technical leadership

Open decisions may proceed as assumptions or development stubs only when they are explicitly approved, have a defined scope, and have a documented condition that prevents them from being mistaken for operational readiness.

17. Definition of Ready for Implementation

The primary control added in Version 2.0 is a Definition of Ready. Implementation must not begin for a feature or delivery increment merely because the product intent is understood. The affected requirement set must be sufficiently decomposed and governed to support correct implementation and verification.

A feature or implementation increment is ready only when:

1. The applicable PRD requirement and intended user/persona are identified.
2. The intended user outcome and scope boundary are understood.
3. Functional behavior and business rules are defined at the level needed for implementation.
4. Required product concepts, states, and relationships are defined.
5. UX treatment is reviewed where user interaction is involved.

6. External dependency decisions are resolved or explicitly classified as approved assumptions or development stubs.
7. Security, access, privacy, and data-handling implications are identified.
8. Acceptance criteria and verification method are defined.
9. The requirement is present in the RTM and linked to the downstream specifications that govern its implementation.
10. No unresolved contradiction exists among the governing specifications.

17.1 Allowed Status for an Unresolved Item

Implementation Blocker — The affected implementation may not proceed until the decision or requirement is resolved.

Approved Assumption — Implementation may proceed using a documented assumption approved by the appropriate owner; the assumption remains traceable and subject to later confirmation.

Approved Development Stub — A temporary non-operational mechanism may be used for development/testing within documented boundaries and must not satisfy operational release criteria.

Deferred Requirement — The item is intentionally outside the current increment or MVP scope and is traceably deferred rather than silently omitted.

A feature is not “done” merely because the UI renders or an endpoint responds. Completion requires satisfaction of the product acceptance condition, applicable functional/technical contracts, and verification evidence.

18. Requested Pivota Starting Action

Use this document as the product contract and perform a discovery pass before implementation. The discovery pass must produce:

1. Confirmed personas and jobs to be done
2. Primary and secondary user journeys
3. Proposed information architecture
4. Initial logical data model consistent with the conceptual model in this PRD
5. MVP feature decomposition mapped to the F1–F9 feature groups and sub-features
6. Key assumptions, open questions, dependencies, blockers, and approved-stub candidates
7. Wireframes for catalog/search, innovation record, executive perspective, technical perspective, opportunity submission, innovation contribution, engagement request, and curator workflows
8. Delivery plan with review gates and proposed implementation increments
9. Traceability from this PRD to proposed functional requirements and acceptance criteria
10. Decision register identifying what must be decided before each implementation increment
11. Proposed functional requirement decomposition and detailed acceptance criteria
12. Dependency/blocker/stub decision register
13. Non-functional and security verification approach
14. Definition-of-Ready assessment for each proposed implementation increment

The first review must focus on whether Pivota has correctly understood the problem, intended product behavior, trust model, user journeys, scope, and acceptance conditions — not on code generation.

No implementation increment should proceed until the associated Definition of Ready has been reviewed and accepted by the appropriate product and technical decision owners.

Pivota must preserve the intent and governance expressed in this PRD through implementation rather than merely producing a functional portal.